

MUHAMMAD AHMED IMTIAZ

Junior AI/ML Engineer

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Summary

Junior AI/ML Engineer with a strong foundation in machine learning and deep learning and hands-on experience delivering reproducible prototypes and production components. Practical experience with PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, RAG/FAISS retrieval, embeddings, XGBoost and graph models (Node2Vec, GraphSAGE). Built and deployed REST inference endpoints using FastAPI/Flask, produced model checkpoints and evaluation reports (AUC/F1), and implemented data preprocessing, feature engineering, experiment tracking. Seeking a junior role to contribute to model development, deployment and performance monitoring.

Technical Skills

- **Primary / Confident:** Python · SQL · Git / GitHub · PyTorch · scikit-learn · FastAPI · Flask · Deep Learning · CNNs (EfficientNet) · Transformers (BERT) · Embeddings · Model evaluation (AUC / F1) · Experiment tracking · Reproducible notebooks · MySQL · OpenCV · ngrok · Docker · Docker Compose · PostgreSQL · Apache Kafka · Go (Golang) · Next.js · Microservices · REST API design · InsightFace · PaddleOCR · Claude Code · Cursor · Windsurf
- **Familiar / Working knowledge:** TensorFlow · Hugging Face · FAISS · RAG · XGBoost · Rasa · n8n · Open-source STT/TTS (whisper.cpp, Vosk, Coqui/Piper) · C++ · Java · Gin framework · GORM · Kafka-Go · React
- **Professional Skills:** Problem-solving · Analytical thinking · Technical communication · Team collaboration · Time management · Adaptability

Experience

AI/ML Intern, Neuronix Technologies

Mar 2026 – Present

- Built a production-ready multi-tenant biometric identity verification platform integrating face matching (InsightFace), liveness detection (FSFM ViT-B16), OCR (PaddleOCR), and fingerprint matching (DeepPrint) as independent FastAPI microservices.
- Designed an event-driven pipeline using Apache Kafka for async result streaming, with a Go consumer persisting structured results to PostgreSQL and a Go/Gin admin backend serving a Next.js dashboard.
- Developed a data acquisition service (Go) and customer-facing portal (Next.js) as single entry points, coordinating parallel AI service calls and storing biometric evidence with full audit trail.
- Containerized and deployed the full 13-service stack using Docker Compose on a Linux VM, resolving production issues including Kafka networking, model memory constraints, and service orchestration.

AI/ML Intern & Team Lead, Codecelix NASTP

Nov 2025 – Feb 2026

- Lead prototyping and delivery of applied ML features; coordinate technical tasks across intern team members.
- Voice Customer Support Agent — Implemented and integrated components of a low-latency voice pipeline (STT → LLM → TTS) using open-source stacks.
- EchoSee features — implemented sound-event detection, 360° directional audio awareness, and an experimental lip-reading captioning prototype.

AI Intern, Dam Techhub

Jul 2025 – Aug, 2025

- Cleaned and preprocessed Amazon co-purchase graph and engineered node features for model training.
- Trained Node2Vec embeddings; developed GraphSAGE link-prediction model (validation AUC 0.845).
- Implemented ASIN mapping and a FastAPI production endpoint (/recommend) with JWT/HTTPBearer protection.
- Delivered reproducible artifacts: node_features.csv, edge_index.pt, model checkpoints and inference scripts.

Fellow (DL/ML Track), Buildables (formerly Bytewise)

Aug 2025 – Nov 2025

- Completed a competitive 12-week mentor-led fellowship focused on DL/ML best practices, experiment tracking, and reproducible pipelines.
- Final project: **Agentic RAG-based Smart Farming Advisor** — hybrid agent integrating FAISS + SentenceTransformers retrieval, a tabular crop predictor (XGBoost/RandomForest) and an EfficientNet-based disease detector; implemented rule-based agent routing and delivered repo, notebooks, evaluation scripts.
- Completed short experiments: ANN vs CNN (MNIST), Cat vs Dog classifier (transfer learning).

Projects

Feedback Flow — Final Year Project (Backend Lead)

2024 – 2025

- Designed Flask APIs, fine-tuned BERT for sentiment analysis, and implemented a Rasa conversational agent integrated with a Flutter frontend (Rasa REST, ngrok for testing).
- Built a FastAPI proxy to authenticate and forward messages between Flutter and Rasa; added in-app visualizations for analytics.

WhatsApp Automation for Small Restaurants — n8n + Gemini

- Built n8n workflows using the Gemini API and Google Sheets for order intake, templated confirmations, and status updates; implemented simple memory and runbook for deployment/testing.

Education

BS in Artificial Intelligence, NUML University, Islamabad

2022 – 2026

FSc (Pre-Engineering)
Askaria College, Rawalpindi

2019 – 2021

Matriculation (Science)
F.G. Sir Syed Boys Secondary School, Rawalpindi

2017 – 2019

Certifications

- Buildables Fellowship (DL/ML Track) — Completion Certificate
- Graphic Designing (CorelDRAW, Photoshop, Illustrator, InGraph) — Axiom Institute
- Data Science Foundations — Online MOOCs